

Question. (Erdős #138 (differences)) Let the van der Waerden number $W(k)$ be such that whenever $N \geq W(k)$ and $\{1, \dots, N\}$ is 2-coloured there must exist a monochromatic k -term arithmetic progression. Is it true that $W(k+1) - W(k) \rightarrow \infty$?

Proof. We will show that $W(k+1) \geq W(k) + k$, which establishes that $W(k+1) - W(k) \rightarrow \infty$. Given a 2-coloring of the first $W(k) - 1$ integers without a monochromatic k -AP, we can extend it by k further elements without creating a monochromatic $(k+1)$ -AP by proceeding greedily. Adding new elements one by one, suppose we have validly colored up to M (where $M < W(k) - 1 + k$); we simply color $M+1$ red if doing so doesn't create a red $(k+1)$ -AP, and blue otherwise.

The only way this algorithm could result in an invalid coloring is if both choices are blocked, meaning there is already a red k -AP with some step size d_R and a blue k -AP with some step size d_B such that the $(k+1)$ -th element for both progressions lands exactly on $M+1$. But this is impossible. Because our original interval up to $W(k)-1$ has no monochromatic k -APs, these progressions must contain at least one newly added element, which bounds their step sizes to $d_R, d_B \leq k-1$. Hence, if we step backward d_B times along the red progression and d_R times along the blue progression, both calculations land exactly on the positive integer $M+1 - d_R d_B$, meaning this single point would have to be simultaneously colored red and blue, which is a contradiction. $\square$